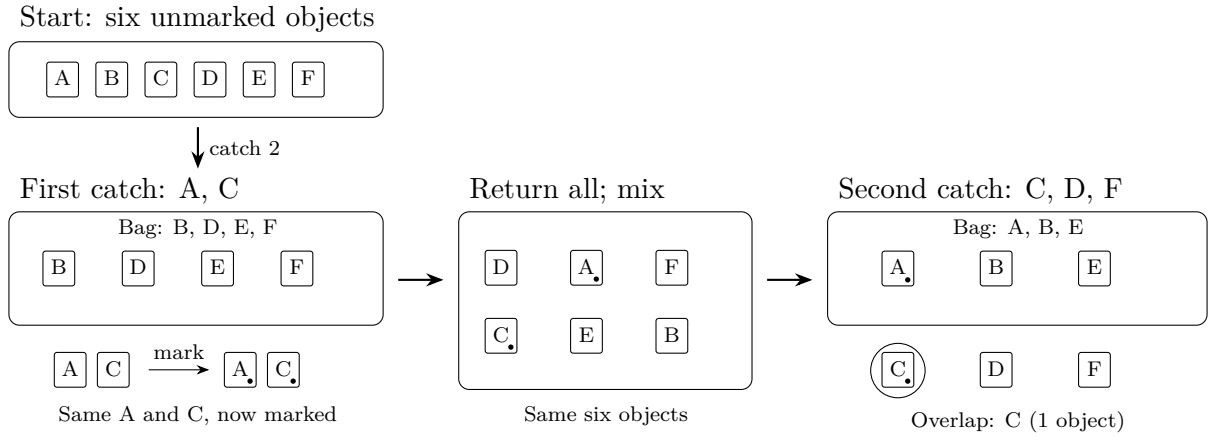
Use identical-size cards or counters with visible marker dots that cannot be felt. Keep the same objects throughout a series; add, remove or exchange none except for temporary catches. For sampling trials, draw without looking; every group of the chosen size should have the same chance, whether marked or unmarked.

Keep each catch out until it is counted. Mark the first catch, return *all* of it, and mix before the second catch. Marks stay on through both catches. An overlap is a marked object in the second catch. Before a new round, return everything and clear the marks. To show a record is possible, you may arrange catches: use distinct objects in each catch, from the same bag.




**Problem 1:** The keeper hides 8 to 20 objects. Catch 4, mark and return them, then catch 4 again. Play three rounds with the same objects. Choose a guess before the keeper reveals the total.

| Round | Overlap | My guess |
|---|---|---|
| 1 | | |
| 2 | | |
| 3 | | |

Keeper's total: _____

**Problem 2:** Build the smallest bag that could give these catches. Build a larger bag that could give exactly the same catches. Which of these totals are possible: 6, 7, 8, 12, 20?

| | | | | | |
|---|---|---|---|---|---|
| First catch | A | B | C | D | E |
| Second catch | B• | D• | E• | F | G |

**Problem 3:** Now the two catches share no objects. Find the smallest possible bag. Is there a largest possible bag if you have as many cards as you want?

| | | | | | |
|---|---|---|---|---|---|
| First catch | A | B | C | D | E |
| Second catch | F | G | H | I | J |

**Problem 4:** These example rounds came from one unchanged bag. It holds either 8 or 16 objects. Every round has a fresh first catch of 4 and a second catch of 4. Which total would you guess from each round alone? Use all six rounds to choose one guess.

| Round | Overlap | Guess: 8 or 16 |
|---|---|---|
| 1 | 2 | |
| 2 | 1 | |
| 3 | 3 | |
| 4 | 1 | |
| 5 | 2 | |
| 6 | 0 | |

My guess from all six rounds: _____